

Example 16.6.1. The managing director of a firm claims that his firm produces 110 items on average daily. A random sample of 15 days gives the following data set.

110, 118, 130, 140, 142, 146, 112, 100, 95, 98, 96, 122, 123, 124, 130.

It is known that the number of items produced by the firm follows normal distribution with variance 300.

Can we conclude at 5% level of significance that the average daily production of items of that firm is
a) 110 items; b) More than 110 items; c) Less than 110 items?
d) Compute p-value for each case.

Solution. (a) Steps involved in testing the hypothesis will be followed in this case:

i) First, we have to formulate null and alternative hypothesis. It is a two-tailed test. Since if the average number of items produced by the firm is more or less than 110 to some extent, then the claim of the managing director will be proved as false. In that case the claim would be rejected.

So, the null hypothesis and alternative hypothesis can be formulated as follows.

Null hypothesis: $H_0: \mu = 110$

Alternative hypothesis: $H_A: \mu \neq 110$.

ii) **Level of significance is:** $\alpha = 0.05$.

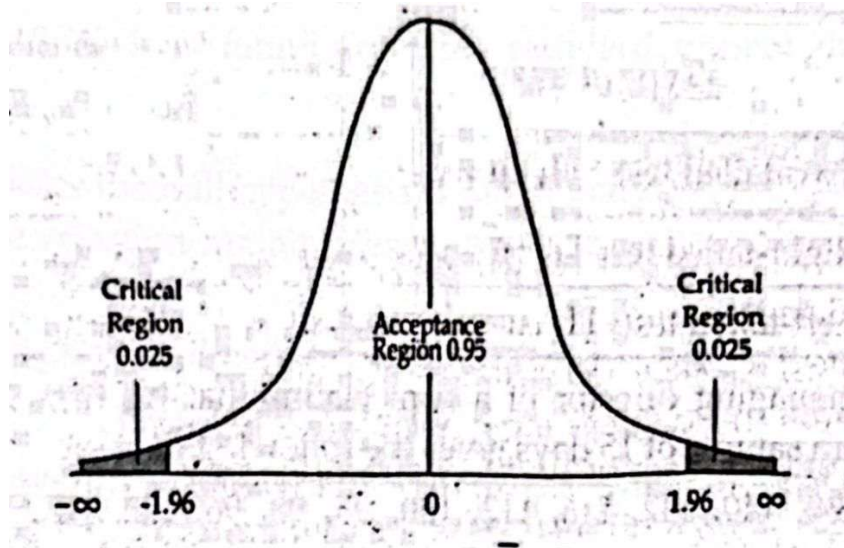
iii) Here, the sample is taken from a normal population with known variance. The appropriate test statistic is

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

where $\bar{X}$ = The sample mean; μ = Population mean (to be tested); σ = Standard deviation of the population; $\sigma/\sqrt{n}$ = Standard error of the sample mean $\bar{X}$.

iv) Here $\alpha = 0.05$ and it is a two tailed-test, the critical region will be on both sides of curve of Z in such a way that the critical region will comprise 2.5% or 0.025 area at the right end and 2.5% at the left end. From the table of area of standard normal distribution, we see that these

values of Z are ± 1.96 , that means the critical regions are $Z < -1.96$ (at the left end) and $Z > 1.96$ (at the right end). Here $|z_{0.025}| = 1.96$.



(v) Under the null hypothesis, the value of Z is: $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$.

(vi) Here, $\bar{x} = \frac{\sum X}{n} = \frac{1786}{15} = 119.07$, $\mu_0 = 110$, $\sigma^2 = 300$, $\sigma = 17.32$ and $n = 15$. Substituting these values in the formula of Z , we have

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{119.07 - 110}{17.32/\sqrt{15}} = 2.03.$$

(vii) It is found that the observed value of Z is 2.03, which is greater than the right tail critical value 1.96, hence it falls in the upper critical region.

(viii) Since the observed value of the test statistic falls in the critical region, so we fail to accept the null hypothesis at 5% level of significance.

- **Conclusion.** Hence, we cannot accept the claim of the managing director at 5% level of significance.
- **p-value.** From the table of the standard normal distribution, we find that $P(Z > 2.03) = 0.0212$, since it is a two tailed test, the p-value is $0.0212 \times 2 = 0.0424$, that means the smallest level of significance at which the hypothesis may be rejected is approximately 4.34%.

(b) Null hypothesis is the same as (a)

Null hypothesis: $H_0 : \mu = 110$;

Alternative hypothesis: $H_1 : \mu > 110$;

Level of significance is: $\alpha = 0.05$.

The appropriate test statistic is the same as (a). That is

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Here $\alpha = 0.05$ and it is one-sided right tailed-test, the critical value $z_{0.05}$ will be found in such a way that $P[Z > z_{0.05}] = 0.05$. It is found from the standard normal distribution that $z_{0.05} = 1.645$.

The calculated value of Z under the null hypothesis is 2.03 which is greater than 1.645. That is observed value of Z lies in the rejection region. Hence, we have no reason to accept the null hypothesis.
p-value. From the table of the standard normal distribution, we find that

$$p = P(Z > 2.03) = 0.0212.$$

That means the smallest level of significance at which the hypothesis may be rejected is approximately 2.12%.

(c) Null hypothesis is the same as (a)

Null hypothesis: $H_0 : \mu = 110$

Alternative hypothesis: $H_1 : \mu < 110$

Level of significance is: $\alpha = 0.05$.

The appropriate test statistic is the same as (a). That is

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Here $\alpha = 0.05$ and it is one-sided left tailed-test, the critical value $z_{0.05}$ will be found in such a way that $P[Z < -z_{0.05}] = 0.05$. It is found from the standard normal distribution that for left tail $z_{0.05} = -1.645$.

The calculated value of Z under the null hypothesis is 2.03 which is greater than -1.645 . That is observed value of Z lies in the acceptance region. Hence, we have no reason to reject the null hypothesis. At 5% level of significance under the alternative hypothesis that the average production of the firm is less than 110.

p-value. From the table of the standard normal distribution, $p = P(Z > 2.03) = 0.0212$, we find that

This is a left tailed test, so the p-value is

$$p = P(Z > -2.03) = 1 - 0.0212 = 0.9788.$$

That means the smallest level of significance at which the hypothesis may be rejected is approximately 97.88%.

Note: In practice, if we are in a position to reject a null hypothesis, we compute p-value to find the exact level of significance. The p-value found in this case is quite unjustified.

2. Population normal, variance unknown and sample size is large ($n > 29$).

Example 16.6.2. Manager of a fertilizer factory claims that the average daily production of his factory follows normal distribution with mean production 880 kg. A random sample of 50 days

shows that average production is 871 kg with standard deviation 21 kg. Test the significance of the claim of the manager at 5% level of significance. Also find p-value.

Solution. Here null and alternative hypotheses are

$$H_0 : \mu = 880 \quad \text{and} \quad H_1 : \mu \neq 880.$$

Here population is normal but variance is unknown and the sample size is large. The sample standard deviation can be taken as a good estimate of the population standard to estimate the standard error of the sample mean $\bar{X}$. That is $\frac{\sigma}{\sqrt{n}}$ can be replaced by $\frac{s}{\sqrt{n}}$ for defining Z . The appropriate test statistic is $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$, which is approximately a standard normal variable. The value of the test statistic Z under the null hypothesis is

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Here s^2 is the estimate of population variance σ^2 , defined as

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2.$$

It is a two-tailed test, so the critical region at 5% level of significance is

$$|Z| > 1.96.$$

We have, $\bar{x} = 871$, $s = 21$, $n = 50$, so the computed value of Z under null hypothesis is

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{871 - 880}{21/\sqrt{50}} = -3.03.$$

- **Decision.** Since the observed value of Z lies in the critical region, so we fail to accept null hypothesis. That means, the manager's claim is not justified.
- **p-value.** From the standard normal integral table, we find that $P(Z > 3.03) = 0.0005$ and $P(Z < -3.03) = 0.0005$ so the value of p is $0.0005 \times 2 = 0.001$. Since the p-value is far less than 0.01, the value of Z is highly significant.

3. Population is not normal, variance known and sample size is large.

Example 16.6.3. The producer of a company claims that the selling price of his product is very standard and it is TK. 1500 per unit with standard deviation TK. 45. There is some doubt of CAB (Consumers' Association of Bangladesh) regarding this price. They want to verify this price using statistical testing procedure. A random sample of the selling prices of 100 products of this company from different areas were collected. The average price per unit was found Tk. 1510.

Can the CAB conclude at 5% level of significance that the average price of the product is standard? Also calculate p-value and 95% confidence interval for population mean.

Solution. We have to test the hypothesis that the sailing price of the product is TK. 1500. So, the null hypothesis and alternative hypotheses are

Null hypothesis: $H_0 : \mu = 1500$;

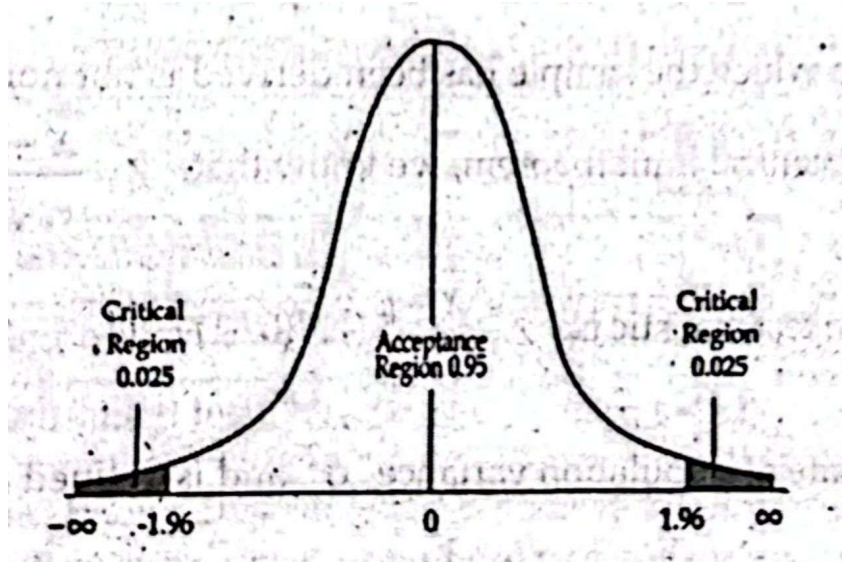
Alternative hypothesis: $H_1 : \mu \neq 1500$;

Level of significance: $\alpha = 0.05$.

Here, we have to test the significance of a population mean with known population variance. But nothing is said about the form of distribution, but mean and variance exists. Since the sample size is large ($n = 100$), according to the central limit theorem, the sampling distribution of the mean is approximately normally distributed with mean μ and standard error $\frac{\sigma}{\sqrt{n}}$, so the appropriate test statistic is:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Here $\alpha = 0.05$ (given), and it is a two tailed-test, the critical region will be on both sides of curve of z in such a way that the critical region will comprise 2.5% or 0.025 area at the right end and 2.5% at the left end. From the table of area of normal curve, we see that these values of Z are ± 1.96 , that means the critical regions are $Z < -1.96$ (at the left end) and $Z > 1.96$ (at the right end).



Under the null hypothesis Z is given by: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

Here, $\bar{x} = 1510$, $\mu_0 = 1500$, $\sigma = 45$ and $n = 100$, then;

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{1510 - 1500}{45/\sqrt{100}} = 2.22.$$

It is found that the observed value of Z is 2.22. It is greater than the critical value 1.96; hence it falls in the upper critical region. So we fail to accept the null hypothesis at 5% level of significance.

Conclusion. The claim of the producer is not right, that means, the price of the products as claimed by the producer is not standard.

- **p-value.** From the table of area under the standard normal distribution we find that the value of $P(Z > 2.22) = 0.0132$ since it is a two tailed test, the p-value is $0.0132 \times 2 = 0.0264$, that means the smallest level of significance at which the hypothesis is rejected is 2.64%.
- **Confidence interval.** 95% confidence interval for μ is given by

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad \text{here } z_{\alpha/2} = z_{0.025} = 1.96.$$

So, the required CI is: $1510 \pm 1.96 \frac{45}{\sqrt{100}} = (1501, 1519)$.

Thus, we may be 95% confident that the true mean price will be between TK. 1501 and TK. 1519.

4. Population is not normal; variance is unknown and the sample size is large.

Example 16.6.4. A manufacturer of fluorescent tubes claims that his tubes have a lifetime of 1950 burning hours. A random sample of 100 tubes is taken from a day's output and tested for burning life. It is found to have a mean burning lifetime of 1900 hours with a standard deviation of 150 hours. Can the claim of the manufacturer be accepted at 5% level of significance? Also find p-value.

Solution. Here nothing is said about population and the population variance is not known.

$$H_0 : \mu = 1900 \quad \text{and} \quad H_1 : \mu \neq 1900.$$

The population from which the sample has been derived is not normal, but the sample size is large. So by the virtue of central limit theorem, we know that: $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0, 1)$.

Thus the appropriate test statistic is: $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0, 1)$.

Here s^2 is the estimate of population variance σ^2 and is defined as

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2.$$

It is a two-tailed test, so the critical region at 5% level of significance is $|Z| > 1.96$.

Given, $\bar{x} = 1900$, $s = 150$, $n = 100$, so the computed value of Z under null hypothesis is

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{1900 - 1950}{150/\sqrt{100}} = -3.33.$$

- **Decision.** Since the observed value of Z lies in the critical region, so we fail to accept null hypothesis. That means, the manufacturer's claim is not accepted at 5% level of significance.
- **p-value.** From the standard normal integral table, we find that $P(Z < -3.33) = 0.0004$ so the value of p is $0.0004 \times 2 = 0.0008$ (approx). Since the p-value is far less than 0.01, the value of z is highly significant.

Population normal, variance unknown and sample size is small ($n < 30$).

Suppose $X_1, X_2, \dots, X_n$ be a random sample of size n drawn independently from a normal population with mean μ and unknown variance σ^2 .

The following null and alternative hypotheses may be considered.

- i) $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ (for a two tailed alternative);
- ii) $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$ (for a right tailed alternative);
- iii) $H_0 : \mu = \mu_0$ against $H_1 : \mu < \mu_0$ (for a left tailed alternative).

The test statistic for testing the null hypothesis H_0 for all the alternatives is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \text{where} \quad s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2.$$

Here t follows Student's t -distribution with $(n-1)$ degrees of freedom.

The decision rules at 100α percent level of significance for hypothesis testing using t -statistic are as follows.

Table 16.5. Decision rule for t -test

Case No.	Type of test	Decision rule: Reject H_0 , if
1	Two-tailed test: $H_1 : \mu \neq \mu_0$	$ t > t_{\alpha/2; (n-1)}$
2	Right-tailed test: $H_1 : \mu > \mu_0$	$t > t_{\alpha; (n-1)}$
3	Left-tailed test: $H_1 : \mu < \mu_0$	$t < -t_{\alpha; (n-1)}$

Example 16.6.5. A wholesaler knows that on average sales in its store is 20% higher in December than in November. For the current year, a random sample of six stores was taken. Their percentage of sales increased in December was found to be 19.2, 18.4, 19.8, 20.2, 20.4, 19.0. Assuming that the sample has been drawn from a normal population with mean μ and unknown variance σ^2 .

- a) test the null hypothesis at 10% level of significance whether the true mean percentage sales increase is 20%, against the two-sided alternative.
- b) do you think that the true mean percentage sales increase is more than 20% at 10% level of significance.
- c) do you think that the true mean percentage sales increase is less than 20% at 10% level of significance.

Solution. Here the population variance is unknown and the sample size is small. The estimated standard error of sample mean $\bar{x}$ is given by

$$se(\bar{x}) = s/\sqrt{n} \quad \text{where, } s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}.$$

(a) We want to test the null hypothesis

$$H_0 : \mu = \mu_0 = 20 \quad \text{against the alternative } H_1 : \mu = \mu_1 \neq 20,$$

Since the sample size is small and population variance is unknown, the value of the test statistic under null hypothesis is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

which is distributed as Student's t with $(n - 1)$ degrees of freedom.

It is a two tailed test, so the decision rule is

$$\text{Reject } H_0 \text{ in favor of } H_1 \text{ if } t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} > t_{\alpha/2; (n-1)} \text{ or, } t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} < -t_{\alpha/2; (n-1)}.$$

The critical values of t at 10% level of significance with $n - 1 = 5$ df are $\pm t_{n-1; \alpha/2} = \pm t_{5; 0.05} = \pm 2.015$ (From table of t -distribution).

Here, $n = 6$, $\bar{x} = 19.5$, $s^2 = 0.588$, and $s/\sqrt{n} = 0.24$.

Then, we have, $t = \frac{19.5 - 20}{0.24} = -1.08$.

Since the observed value $t = -1.08$ lies between -2.015 and 2.015 , hence we fail to reject the null hypothesis at 10% level of significance.

That means, the true mean sales is 20% higher in December than in November.

(b) In this case, we have to perform a one-tailed test, given by

$$H_0 : \mu = \mu_0 = 20 \quad \text{against the alternative } H_1 : \mu = \mu_1 > 20.$$

The decision rule is to reject H_0 in favor of H_1 if $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} > t_{\alpha; n-1}$.

The computed value of t is the same as before i.e., $t = -1.08$, the critical value of t at 10% level of significance is: $t_{0.10, 5} = 1.476$.

Since the observed value of $t = -1.08$ which is less than the critical value, we fail to reject the null hypothesis at 10% level of significance, which means the average sales increased by more than 20 percent is not evident from the given data.

(c) In this case, we have to perform a one-tailed test, given by

$$H_0 : \mu = \mu_0 = 20 \quad \text{against the alternative } H_1 : \mu = \mu_1 < 20.$$

The decision rule is to reject H_0 in favor of H_1 if $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} < -t_{\alpha; n-1}$.

The computed value of t is the same as before i.e., $t = -1.08$, the critical value of t at 10% level of significance is: $t_{0.10, 5} = -1.476$.

Since the observed value of $t = -1.08$ lies beyond the critical region, we fail to reject the null hypothesis at 10% level of significance, which means the average sales increased by less than 20 percent is not evident from the given data.

Example 16.6.6. (Large sample with unknown variance when parent population is not normal) A fertilizer factory manager claims that its average daily production is 912 kg. A random sample of 50 days shows that average production is 903 kg with standard deviation 21 kg. Test the significance of the claim of the manager at 5% level of significance.

Solution. Here, $H_0 : \mu = 912$ and $H_1 : \mu \neq 912$.

The population from which the sample has been derived is not normal, but the sample size is large. So by the virtue of central limit theorem, we know that $\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0, 1)$.

Thus under the null hypothesis, the appropriate test statistic is

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

where s^2 is the estimate of population variance σ^2 , defined as $s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$.

It is a two tailed test, so the critical region at 5% level of significance is $|Z| > 1.96$.

Given, $\bar{x} = 903$, $s = 21$, $n = 50$, so the computed value of Z under null hypothesis is

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{903 - 912}{21/\sqrt{50}} = -3.03.$$

Decision. Since the observed value of Z lies in the critical region, so we fail to accept null hypothesis. That means, the manager's claim is not justified.

Example 16.6.7. (Two-tailed test for small sample with known variance) The yields of wheat on a random sample of six test plots are as 1.40, 1.80, 1.30, 1.90, 1.60 and 2.20 tons per acre, test whether the information supports the claim that the average yield for this kind of wheat is 1.5 tons/acre with standard deviation 0.43 tons/acre. Also find the p-value.

Solution. The null and alternative hypotheses for this test are

$$\text{Null hypothesis: } H_0 : \mu = 1.5$$

$$\text{Alternative hypothesis: } H_A : \mu \neq 1.5.$$

Here, the sample size is small, the population variance is known and the sample is taken from normal population, so the appropriate test statistic is

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Let the level of significance $\alpha = 0.05$ (if α is not given, it is considered as 0.05). Thus, the critical values of Z are ± 1.96 that means the critical regions are $Z < -1.96$ and $Z > 1.96$.

The average yield as calculated for the observations is 1.7 tons/acre, so the value of Z under null hypothesis is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{1.7 - 1.5}{0.43/\sqrt{6}} = 1.14.$$

Since the observed value of Z is less than 1.96, that means the observed value lies in the acceptance region, so we fail to reject null hypothesis. Hence, the information supports that the average yield is 1.5 tons/acre.

- **p-value.** From the table of standard normal distribution, we find that $P(Z > 1.14) = 0.1271$ and $P(Z < -1.14) = 0.1271$ so, the p-value is $0.1271 \times 2 = 0.2542$ that means the test will be significance at 25.42% level of significance.

Example 16.6.8. (Right tailed test for small sample size with known variance) A stenographer claims that she can take dictation at the rate of more than 100 words per minute with a standard deviation of 15 words. Can we reject the claim on the basis of 10 trials in which she demonstrates a mean of 102 words per minute? Use 5% level of significance.

Solution. If the stenographer can take dictation of even at the rate of 100 words per minute, her claim can not be accepted. So, the null hypothesis and alternative hypothesis to be considered are

$$\begin{aligned}\text{Null hypothesis: } & H_0 : \mu = 100 \\ \text{Alternative hypothesis: } & H_1 : \mu > 100.\end{aligned}$$

Since the population variance is known, the appropriate test statistic is $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$.

The critical region at 5% level of significance is: $Z > 1.645$. Here, $\bar{x} = 102$, $\sigma = 15$, $n = 10$, so the value of Z is

$$Z = \frac{102 - 100}{15/\sqrt{10}} = 0.42.$$

The computed value of Z does not exceed the critical value 1.645, so we fail to reject null hypothesis. So, the stenographer's claim is not correct.

Example 16.6.9. (Left tailed test for small sample size with known variance) An automobile manufacturer company claims that a new model car achieves on an average 31.5 miles per gallon in highway driving. The distribution is known to be normal with standard deviation 2.4 miles per gallon. A random sample of sixteen automobiles provided an average of 30.6 miles per gallon in highway trials. Test the claim of company at the 5% level of significance against the population mean is less than 31.5 miles per gallon.

Solution. It is a left-tailed test, because, if the average coverage of distance is even equal to 31.5 miles, the company's claim will not be correct. Thus, the null and alternative hypotheses are

$$\text{Null hypothesis: } H_0 : \mu = 31.5$$

Alternative hypothesis: $H_A : \mu < 31.5$.

Since, the population variance is known, the appropriate test statistic is $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$.

The critical region at 5% level of significance is: $Z < -1.645$. Here, $\bar{x} = 30.6$, $\sigma = 2.4$, $n = 16$, so the value of Z is

$$Z = \frac{30.6 - 31.5}{2.4/\sqrt{16}} = -1.50.$$

The computed value of Z does not fall in the critical region, so we fail to reject the null hypothesis that means, the average achievement of car is not less than 31.5 miles per gallon.

Example 16.6.10. (Two-tailed test for large sample with known variance) A large manufacturer of stereo components is concerned about the efficiency of many new employees hired during the last six months. The efficiency rating of all employees has been reasonably stable with mean rating 200 and a standard deviation of 20. A random sample of 75 new employees has been selected and their average efficiency rating is found as 197.5. Test the null hypothesis at 1% level of significance that the mean efficiency rating is still 200. Find 99% confidence interval for population mean.

Solution. It is a two-tailed test because if the mean efficiency rating obtained from sample is too high or too low in comparison with population mean rating, the null hypothesis would be rejected. The null and alternative hypotheses to be considered are

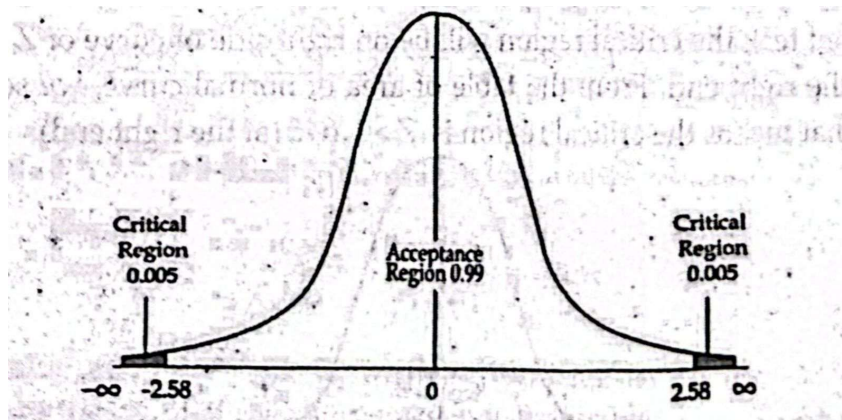
Null hypothesis: $H_0 : \mu = 200$

Alternative hypothesis: $H_1 : \mu \neq 200$.

We have to test the significance of a population mean with known population variance σ^2 , and the sample size is also large, so the sampling distribution of the mean is normally distributed with standard error $\frac{\sigma}{\sqrt{n}}$, the appropriate test statistic for the selected hypothesis is

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Here the level of significance is $\alpha = 0.01$ (as given in the problem).



It is a two-tailed test, the critical region will be on both ends of curve of Z that will comprise of 0.005 area at the right end and 0.005 area at the left end. From the table of normal curve,

we see that the critical values of Z are ± 2.58 , that means the critical regions are $Z < -2.58$, $Z > 2.58$.

Now, we have to calculate:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Here, $\bar{x} = 197.5$, $\mu_0 = 200$, $\sigma = 20$ and $n = 75$, we have

$$z = \frac{197.5 - 200}{20/\sqrt{75}} = -1.08.$$

The observed value of Z is greater than the lower critical value -2.58 . Since the observed value of test statistic does not fall within the critical region, so we fail to reject the null hypothesis at 1% level of significance. That means the mean efficiency rating is still 200.

- **Confidence interval.** 99% confidence limits are given by

$$\bar{x} \pm z_{0.005} \frac{\sigma}{\sqrt{n}} = 197.5 \pm 2.58 \frac{20}{\sqrt{75}} = (191, 203),$$

that means, we are 99% confident that true average efficiency rating of all employees will be between 191 and 203.

Example 16.6.11. Suppose in Example 16.6.3, instead of checking if the price is different from standard price, CAB decided to verify whether the specified price of the product is more than standard price.

Solution. It is obviously a one-tailed test, because, if average price obtained from the sample is less than or equal to the standard price, the claim of the producer will be established. Hence, we have to consider a composite hypothesis defined as

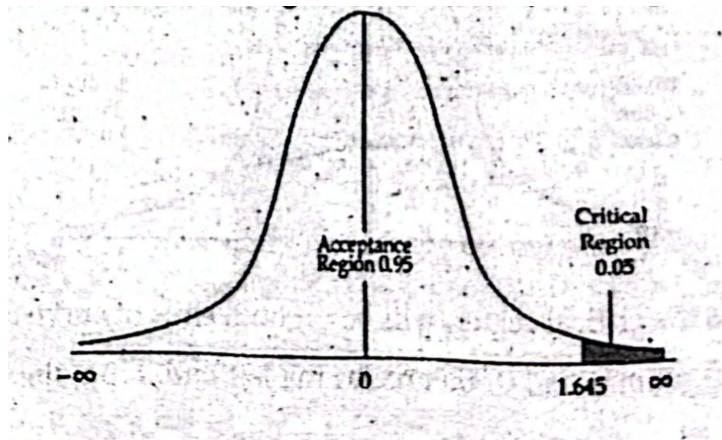
Null hypothesis: $H_0 : \mu \leq 1500$

Alternative hypothesis: $H_1 : \mu > 1500$.

The test statistic for testing the null hypothesis is: $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$.

Here $\alpha = 0.05$ (given).

It is a right-tailed test; the critical region will be on right side of curve of Z that will comprise 5% or 0.05 area at the right end. From the table of area of normal curve, we see that this critical value of Z is 1.645 that means the critical region is $Z > 1.645$ (at the right end).



We have found $z = 2.22$ (from Example 16.6.3).

Hence, the observed value of $Z = 2.22$ which is greater than the critical value 1.645, hence, observed value of test statistic falls in the critical region, we fail to accept the null hypothesis at 5% level of significance.

The claim of the producer is not right, that means, the average price of the products of the producer is more than the standard price.

p-value. The p-value is given by $P(Z > 2.22) = 0.0132$ that means the smallest level of significance at which the hypothesis is rejected is 0.0132 or 1.3%.

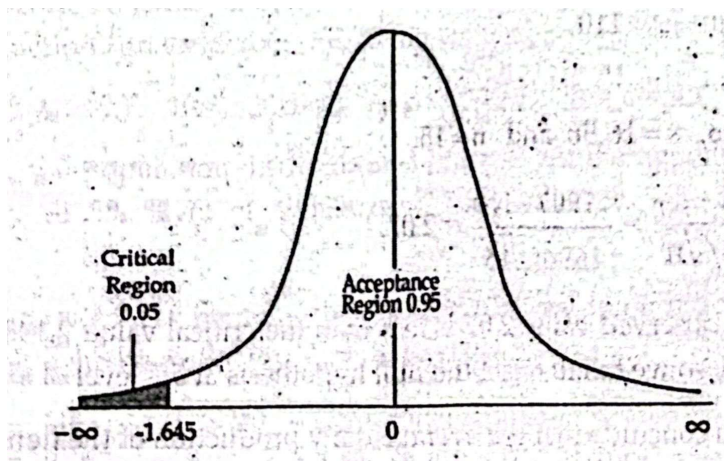
Example 16.6.12. The average petrol consumption of existing auto engines is 10.5 km. per liter. An auto company decided to introduce a new six-cylinder car whose mean petrol consumption is claimed to be lower than that of the existing auto engine. In order to verify company's claim, a sample of 50 new cars was randomly selected and it was found that the mean petrol consumption is 10 km. per liter with a standard deviation of 3.5 km. per liter. Test whether the claim of the company is acceptable at 5% level of significance.

Solution. Here we have to consider the following hypotheses

$$H_0 : \mu = 10.5 \quad \text{and} \quad H_1 : \mu < 10.5.$$

Although the population variance is not known, since sample size is large, the appropriate test statistic is $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ where s^2 is sample variance and n is the sample size.

Since it is a left tailed test, the critical region lies in the left end of the curve given by $Z < -1.645$.



Given, $\bar{x} = 10$ km., $s = 3.5$, $n = 50$, so the value of test statistic Z under H_0 is given by

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{10 - 10.5}{3.5/\sqrt{50}} = -1.01.$$

This observed value of Z does not lie in the critical region, so we fail to reject the null hypothesis. That means the claim of the company is not acceptable.

Example 16.6.13. Suppose the daily number of items produced by a firm for randomly selected 15 days is as follows

110, 118, 130, 140, 142, 146, 112, 100, 95, 98, 96, 122, 123, 124, 130.

Can we conclude at 5% level of significance that the average daily production of items of the firm is 110?

Solution. It is a two-tailed test because if the average hourly number of items produced by the company is more or less than 110, then the statement that the average daily production of items will be proved as false, then the null hypothesis that the average daily production of items is 110 would be rejected.

So, the null hypothesis and alternative hypothesis are as follows

Null hypothesis: $H_0 : \mu = 110$

Alternative hypothesis: $H_1 : \mu \neq 110$.

Level of significance as given in the problem is $\alpha = 0.05$, since the variance is unknown and sample size is small, the appropriate test statistic is

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \quad \text{which is distributed as Student's } t \text{ with } n - 1 \text{ df.}$$

Here, $n = 15$, so the $\text{df} = 15 - 1 = 14$.

Here $\alpha = 0.05$, and it is a two tailed-test, the critical region will be on both sides of t -distribution, in such a way that the critical region will comprise 2.5% or 0.025 area at the right end and 2.5% at the left end. From the table of t -distribution, we find for $\text{df} = 14$, $\alpha = 0.025$, the values of t are ± 2.145 , that means the critical regions are $t < -2.145$ (at the left end) and $t > 2.145$ (at the right end).

Here, $\bar{x} = 119.07$ and $\mu = \mu_0 = 110$.

$\sum(x - \bar{x})^2 = 287.78$, so, $s = 16.96$ and $n = 15$.

So, we have, $t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{119.07 - 110}{16.96/\sqrt{15}} = 2.07$.

It is found that the observed value 2.07 is less than the critical value 2.145; hence it does not fall in the critical region, so we fail to reject the null hypothesis at 5% level of significance.

- **Conclusion.** We can conclude that the average daily production of the items of the given firm can be accepted as 110.

Example 16.6.14. (Small sample with unknown variance) A gas station repair shop claims that the average time it takes to do a lubrication job and oil change is maximum 30 minutes. A consumer protection department wants to test the claim. A sample of six cars was sent to the station for oil change and lubrication. The job took an average of 34 minutes with a standard deviation of 4 minutes. Assuming that the population is normal, do you think that the job takes average of time more than 30 minutes? (use $\alpha = 0.05$)

Solution: It is obviously a one-tailed test, because, the claim will not be rejected if the average time taken on a car for the oil change and lubrication job is considerably less than 30 minutes. So, in this case we have to consider a composite hypothesis given by (although in practice we use simple hypothesis)

$$H_0 : \mu \leq 30 \quad \text{against the alternative} \quad H_1 : \mu > 30.$$

Since the sample size is small and population variance is unknown, t -statistic will be used. Under the null hypothesis the value of t is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \text{which is distributed as Student's } t \text{ with } n - 1 \text{ df.}$$

Here, $n = 6$, $\bar{x} = 34$, $s = 4$ and $\hat{\sigma}(\bar{x}) = s/\sqrt{n} = 1.63$.

Therefore, we get: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{34 - 30}{1.63} = 2.45$.

The critical value of t at $\alpha = 0.05$ and $\text{df} = n - 1 = 5$, for a right tailed test is given by $t_{0.05} = 2.02$, since the computed value of $t = 2.45$ is higher than the critical value of $t = 2.02$, we reject the null hypothesis, that means the claim of the shop is not considered to be correct.

p-value: In this case p-value is given by $P(t_5 > 2.45) = 0.03$ (from the table of critical value of t -distribution), hence p-value is 0.03 or 3%.

Example 16.6.15. A process that produces bottles of shampoo, when operating correctly, produces bottles whose contents weigh, on average, 20 ounces. A random sample of nine bottles from a single production run yielded the following weights

21.4, 19.7, 19.7, 20.6, 20.8, 20.1, 19.7, 20.3, 20.9.

Assuming that the population distribution is normal, test the hypothesis that the process is operating correctly at 5% level of significance. Also calculate 95% confidence interval for population mean.

Solution. It is a two-tailed test, because, the claim will be rejected if the average weight deviates from 20 ounces in any direction. So, the hypothesis is given by

$$H_0 : \mu = 20 \quad \text{against the alternative} \quad H_1 : \mu \neq 20.$$

Since the sample size is small and population variance is unknown, t -statistic will be used. Under the null hypothesis the value of t is

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \quad \text{which is distributed as Student's } t \text{ with } n - 1 \text{ df.}$$

Here, $n = 9$, $\bar{x} = 20.36$, $\mu_0 = 20$, $s = 0.61$ and $\hat{\sigma}(\bar{x}) = s/\sqrt{n} = 0.203$.

Hence, $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{20.36 - 20}{0.203} = 1.77$.

The critical values of t at $\alpha = 0.05$ with $n-1 = 8$ df for a two-tailed test are given by $\pm t_{0.025} = \pm 2.316$.

Since the computed value of t is 1.77 which does not fall in the critical region, so we fail to reject the null hypothesis, that means the process is operating correctly.

95% confidence interval for population mean μ is given by

$$\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} = 20.36 \pm 2.316 \times \frac{0.61}{\sqrt{9}}.$$

So, the lower and upper confidence limits are 19.89 and 20.83 respectively.

16.7. Test of Hypothesis Concerning Two Population Means

Suppose $X_{11}, X_{12}, \dots, X_{1n_1}$ be a random sample of size n_1 drawn from normal population with mean μ_1 and variance σ_1^2 , and $X_{21}, X_{22}, \dots, X_{2n_2}$ be another sample of size n_2 drawn from normal population with mean μ_2 and variance σ_2^2 . Suppose, the observed sample means are $\bar{X}_1$ and $\bar{X}_2$. In the earlier chapter, it is mentioned that the distribution of the difference between two sample means follows normal distribution when variances are known for all possible sample sizes, that means

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1).$$

The possible null and alternative hypotheses considered for testing the significance of difference between two population means are

- i) $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$ (for a two tailed alternative);
- ii) $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 > \mu_2$ (for a right tailed alternative);
- iii) $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 < \mu_2$ (for a left tailed alternative).

Again, the following alternative situations may arise in testing the above mentioned null and alternative hypotheses

- i) Independent samples with known population variances, sample sizes are large or small.
- ii) Independent samples with unknown population variances, sample sizes are large.
- iii) Independent populations for small sample sizes (≤ 29) with unknown but equal variances.
- iv) Independent populations for small sample sizes (≤ 29) with unknown and unequal variances.
- v) Correlated sample or matched sample (the sample obtained from a bi-variate normal population or paired observations).

The test statistic to be used for testing the simple hypothesis $H_0 : \mu_1 - \mu_2 = 0$ against a one-sided or two-tailed alternative is given by

Under situation (i):

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1).$$

Under situation (ii):

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0, 1),$$

where $s_1^2 = \frac{\sum(x_1 - \bar{x}_1)^2}{n_1 - 1}$ and $s_2^2 = \frac{\sum(x_2 - \bar{x}_2)^2}{n_2 - 1}$.

Under situation (iii):

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which is distributed as Student's t with $n_1 + n_2 - 2$ degrees of freedom, where, s^2 is pooled estimate of variance, given by

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

The value of $(\mu_1 - \mu_2) = 0$ for all Z and t defined above under null hypothesis $H_0 : \mu_1 - \mu_2 = 0$, but if any other value is specified (say, $\mu_1 - \mu_2 = \theta_0$) by null hypothesis for the difference between means, that means if $H_0 : \mu_1 - \mu_2 = \theta_0$, then the value of $(\mu_1 - \mu_2)$ will be θ_0 instead of zero.

For testing the hypothesis regarding difference between two means under situation (i) to (iii) it is desirable to test the equality of two population variance to check if the assumptions of equal variances are valid or not. If the variances are found not to be equal, then Student's t -statistic can not be applied. Hence, under situation (iv), under null hypothesis, the test statistic is given by:

$$t' = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

Here, t' is not a Student's t -statistic. The critical values of t' at $100\alpha\%$ level of significance is computed using the formula:

$$t'_\alpha = \frac{\frac{s_1^2 t_1}{n_1} + \frac{s_2^2 t_2}{n_2}}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t_1 and t_2 are Student's t with $(n_1 - 1)$ and $(n_2 - 1)$ df respectively at $100\alpha\%$ level of significance.

Again, under situation (v), let us consider a random sample of n matched paired observations (x_i, y_i) from a bi-variate normal population, then the test of the hypothesis $H_0 : \mu_1 - \mu_2 = 0$ requires to compute a statistic d defined as $d = x_i - y_i$, and the testing procedure is the same as testing the significance of a single mean of the observations obtained from the difference of two variables assuming that small sample size and unknown population variance. The statistic used for testing this type of hypothesis is called paired- t test, defined as

$$t = \frac{\bar{d}}{se(\bar{d})} \sim t_{n-1} \quad \text{under situation (v)}$$

which is distributed as t with $n - 1$ df.

$$\text{where, } \bar{d} = \frac{\sum d}{n}, \quad s_d^2 = \frac{\sum (d - \bar{d})^2}{n - 1} \quad \text{and} \quad se(\bar{d}) = \frac{s_d}{\sqrt{n}}.$$

- **Decision rule.** The decision rules in all of the above cases are the same as that of testing significance of a single mean using Z or t -statistic.

Example 16.7.1. It is wanted to investigate if male and female typists earn comparable wages. A sample data for daily wages of male and female provide with the following information.

Table 16.6. Sample mean and variance of male and female typists

	Male	Female
Sample size	60	60
Mean wage	Taka 158.50	Taka 141.60
SD (Population)	Taka 18.20	Taka 20.60

Test whether the mean wages of male typists is more than female typists at 5% and 1% level of significance.

Solution. Let the wages of male and female are normally and independently distributed with means μ_1 and μ_2 , and known variances σ_1^2 and σ_2^2 respectively. It is a one-tailed test so we consider the following hypothesis

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 > \mu_2.$$

As the sample sizes are large, under null hypothesis, the value of test statistic is given by

$$Z = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

For $\alpha = 0.05$, the critical region is $Z > 1.645$, and for $\alpha = 0.01$, the critical region is $Z > 2.33$.

Here, $\bar{X}_1 = 158.50$, $\bar{X}_2 = 141.6$, $\sigma_1 = 18.20$, $\sigma_2 = 20.60$, $n_1 = n_2 = 60$.

Thus, the computed value of Z is:

$$z = \frac{(158.50 - 141.6)}{\sqrt{\frac{(18.20)^2}{60} + \frac{(20.60)^2}{60}}} = 4.76.$$

Conclusion. The computed value of Z is greater than critical values at both the level of significance, hence in the given city, male typists have on the average higher earnings than their female counterpart at 1% and 5% levels of significance. Here, the value z is highly significant.

Example 16.7.2. (Large sample sizes with unknown population variances) A potential buyer of electric bulbs bought 100 bulbs each of two famous brands A and B. Upon testing both these sample, he found that brand A had a mean life of 1500 hours with standard deviation of 50 hours whereas brand B had an average life of 1530 hours with standard deviation of 60 hours. Can it be concluded at 5% level of significance that the bulbs of two brands differ significantly in quality?

Solution. We assume that the parent population of these two lifetimes are independently distributed with means μ_1 and μ_2 and unknown variances σ_1^2 and σ_2^2 . We also assume that there is no significant difference in the quality of both brands so that brand A is as good as brand B in terms of operating hours. Hence, we have to test

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_2.$$

Since sample sizes are large, so under the null hypothesis, the value of test statistic is

$$Z = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0, 1).$$

Again, $\alpha = 0.05$ and since it is a two tailed test, the critical region is $|Z| > 1.96$.

Here, $\bar{X}_1 = 1500$, $\bar{X}_2 = 1530$, $s_1 = 50$ and $s_2 = 60$, $n_1 = n_2 = 100$.

Now,

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{(50)^2}{100} + \frac{(60)^2}{100}} = 7.81.$$

Thus,

$$z = \frac{(1500 - 1530)}{7.81} = -3.841.$$

Since the observed value of Z is greater than the critical value, the null hypothesis may be rejected at 5% level of significance.

- **p-value.** From the table of Z , we have $P(Z < -3.00) = P(Z > 3.00) = 0.00$, hence the null hypothesis may be rejected even at 0% level of significance, so $p = 0.00$. It is said that the value z is highly significant since the p-value is 0.00.
- **Confidence Interval.** 95% confidence limits for the difference of two population means $(\mu_1 - \mu_2)$ are given by

$$(\bar{X}_1 - \bar{X}_2) \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = (-30) \pm 7.81 = (22.19, 37.81).$$

Example 16.7.3. (Large sample sizes with unknown population variances) A professor taught two sections of an introductory marketing course using very different styles. In the first section approach was extremely formal and rigid, while in the second section an independent, more relaxed and informal attitude was adopted. At the end of the course, a common final examination was administered. In the first section the seventy-two students obtained a mean score of 71.03 and the sample standard deviation was 22.91. In the second section there were sixty-four students, with mean score 80.92 and standard deviation 23.11. Assume that these two groups of students can be regarded as independent random samples from the populations of all students who might be exposed to these teaching methods. Test at 5% level of significance (i) whether the performance of these two methods is the same, (ii) whether second method is better than first one.

Solution. We assume that the parent populations are independently distributed with means μ_1 and μ_2 and unknown variances σ_1^2 and σ_2^2 .

(i) We have to test that there is no significant difference in two method, so that first method is as good as second method as far as the teaching system is concerned. Hence, we have to test

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_2.$$

Since sample sizes are large, so under null hypothesis the value of test statistic is

$$Z = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0, 1).$$

Again, $\alpha = 0.05$ and since it is a two tailed test, the critical region is $|Z| > 1.96$.

Here, $\bar{X}_1 = 71.03$, $\bar{X}_2 = 80.92$, $s_1 = 22.91$ and $s_2 = 23.11$, $n_1 = 72$, $n_2 = 64$.

Now,

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{(22.91)^2}{72} + \frac{(23.11)^2}{64}} = 3.95.$$

Thus,

$$z = \frac{(71.03 - 80.92)}{3.95} = -2.50.$$

Since the computed absolute value of Z is greater than the critical value, the null hypothesis may be rejected at 5% level of significance. The sample supports that there is significant difference between average performances of the two methods.

- **p-value.** From the table of Z , we have $P(Z < -2.50) = P(Z > 2.50) = 0.007$, hence the null hypothesis may be rejected even at 1.4% (since $0.007 \times 2 = 0.014$) level of significance, so $p = 0.014$.
- **Confidence Interval.** 95% confidence limits for the difference of two population means $(\mu_1 - \mu_2)$ are given by

$$(\bar{X}_1 - \bar{X}_2) \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = (-9.89) \pm 7.81 = (-17.70, 2.08).$$

(ii) Since the second method would be proved to be better than the first method, if the average score obtained by first method is significantly smaller than that of second method, so, in order to test whether second method is better than the first method, we have to consider a one-tailed test defined as

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 < \mu_2.$$

Here, since the sample sizes are large, the test statistic is also Z as defined in (i), but the critical region will cover the 5% area only in the left tail, thus the critical region is $Z < -1.645$.

The observed value of Z is -2.50 , as found in (i), falls in the critical region, so the null hypothesis may be rejected at 5% level of significance. That means, the sample supports that the second method is on an average better than the first method.

Example 16.7.4. (Large sample sizes with known variances) A firm believes that the tires produced by process I on an average last longer than tires produced by process II. To test this belief, random samples of tires produced by two processes were tested and the results are as

Table 16.7. Mean and standard deviation of lifetimes of tires

	Process I	Process II
Sample size	50	50
Average lifetime (in km.)	22,400	21,800
Population Standard deviation (in km.)	1000	1000

Is there any evidence at 5% level of significance that the firm is correct in its belief?

Solution. We have to consider the following null and alternative hypothesis

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 > \mu_2.$$

Since the sample sizes are large, the value of test statistic under null hypothesis is

$$z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

Since it is a one tailed test, the critical region is: $Z > 1.645$.

Given, $\bar{x}_1 = 22400$, $\bar{x}_2 = 21800$, $\sigma_1 = 1000$, $\sigma_2 = 1000$, $n_1 = n_2 = 50$.

So, the value of Z is:

$$z = \frac{(22400 - 21800)}{\sqrt{\frac{(1000)^2}{50} + \frac{(1000)^2}{50}}} = 3.00.$$

Since the calculated value of Z is more than its critical value at 5% level, therefore null hypothesis may be rejected. Hence, we can conclude that the tires produced by process I have longer life than process II.

- **p-value.** From the table of area under the standard normal probability distribution, we find $P(Z > 3.00) = 0.0013$ (so the smallest critical value for which null hypothesis may be rejected is 0.001, thus the p-value is 0.001).

Example 16.7.5. (Small sample sizes with unknown and equal population variance) Manager of factory I claims that the average wage of its workers is higher than that of factory II. A survey was conducted on daily wages of workers of two factories to see if the claim of ...

Table 16.8. Mean and standard deviation of wages of workers

	Factory I	Factory II
Sample size	16	11
Sample mean wage	Taka 290	Taka 250
Sample standard deviation	15	50

Solution. Let us assume that the wages of corresponding population of two factories are independent and normally distributed with common unknown variance σ^2 . Thus, the null and alternative hypothesis are

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 > \mu_2.$$

Since the sample sizes are small and variances are unknown but equal, the test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which is distributed as Student's t with $n_1 + n_2 - 2$ degrees of freedom,

where, $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$ is an estimate of common variance σ^2 .

Here, $n_1 = 16$, $n_2 = 11$, so the degrees of freedom is $n_1 + n_2 - 2 = 25$.

It is a one tailed test, thus from the table of t -distribution we have the critical values of t at 5% level of significance with 25 df is 1.708, that means, $t_{25, 0.05} = 1.708$, in other words, the critical region is $t > 1.708$.

Given, $\bar{x}_1 = 290$, $\bar{x}_2 = 250$, $s_1 = 15$, $s_2 = 50$, $s^2 = 1135$ and $s = 33.69$.

So, under the null hypothesis, the value of t is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{290 - 250}{13.19} = 3.03.$$

- **Decision.** The observed value of t is greater than the critical value, it lies in the critical region, so the null hypothesis may be rejected at 5% level of significance.
- **Conclusion.** The claim of manager of factory-I is justified.

Example 16.7.6. (Small sample sizes with unknown and equal population variances) The residents of Dhaka city complains that traffic speeding fines given in their city are higher than the traffic speeding fines that are given in Chittagong city. The appropriate authority agreed to study the problem. To check if the complaints were reasonable, independent random samples of the amounts paid by the residents for speeding fines in each of two cities over the last three months were obtained and shown in following table.

Table 16.9. Amounts of traffic speeding fines

Dhaka city	100	125	135	128	140	142	128	137	156	142
Chittagong city	95	87	100	75	110	105	85	95		

Assuming an equal population variance, test

- whether there is any significant difference in the mean cost of speeding in these two cities and find the 95% confidence interval.
- whether the mean speeding cost in Dhaka city is higher than Chittagong city at 1% level of significance.

Solution. (i) Let X_1 be the speeding cost in Dhaka city and X_2 be the speeding cost in Chittagong city. Assuming the samples have been drawn independently from two normal populations with means μ_1 and μ_2 respectively with common variance σ^2 . We have to test the following hypothesis

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_2.$$

It is a two tailed test, the sample sizes are small and variances are also unknown, so, the appropriate test statistic is given by

$$t = \frac{\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which is distributed as Student's t with $n_1 + n_2 - 2$ degrees of freedom, where $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$

is an estimate of common variance σ^2 .

Given, $n_1 = 10$, $n_2 = 8$, so the degrees of freedom is: $n_1 + n_2 - 2 = 16$.

From the table of t -distribution we have the critical values of t at 5% level of significance with 16 df are ± 2.12 , that means, $t_{16, 0.025} = 2.12$, in other words, the critical region is $|t| > 2.12$.

From the given observations, we have,

$$\bar{x}_1 = 133.30, \quad \bar{x}_2 = 94.00, \quad s_1 = 18.20, \quad s_2 = 20.60, \quad s^2 = 371.98.$$

So, under null hypothesis, the calculated value of t is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{133.30 - 94.00}{9.14} = 4.30.$$

- **Decision.** The observed absolute value of t is greater than the critical value, it lies in critical region, so the null hypothesis may be rejected at 5% level of significance.
- **Conclusion.** There is significant difference between the average traffic fines in two cities.

Again, the 95% confidence interval for $(\mu_1 - \mu_2)$ is given by

$$(\bar{x}_1 - \bar{x}_2) - t_{16, 0.025} \times s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < (\mu_1 - \mu_2) < (\bar{x}_1 - \bar{x}_2) + t_{16, 0.025} \times s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\text{or, } 39.30 \pm 2.12 \times 4.30 = 39.30 \pm 9.12 = (30.18, 48.42).$$

- **p-value.** From the table of critical values of t -distribution, we find that $t_{16, 0.005} = 2.921$, means $2 \times 0.005 = 0.010 = 1\%$ can be considered as the smallest level of significance at which the null is still rejected, so the required p-value is 0.01.

(ii) In this case we have to perform a one-tailed test given by

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 > \mu_2.$$

Here the critical value of t at 1% level of significance with 16 df is: $t_{16, 0.01} = 2.583$.

The observed value of t is same as previous one, so $t = 4.30$.

In this case too, the observed t falls in the critical region, so the null hypothesis may be rejected at 1% level of significance. We can conclude that on an average the traffic fines at Dhaka city is higher than that of Chittagong city.

- **p-value.** In this case the p-value is 0.005 because, $P(t > 4.30) = 0.005$.

Example 16.7.7. (Matched observations or paired sample) A study was conducted by pharmaceutical company to compare the difference in effectiveness of two particular drug ...

cholesterol levels. The company used paired sample approach to control variation in reduction might be due to factors other than the drug itself. Each member of a pair was matched by weight, lifestyle, and other pertinent factors. Drug X was given to one person randomly selected from each pair, and drug Y was given to the other individual in the pair. After a specific period of time each person's cholesterol level was measured again. Suppose a random sample of 8 pairs of patients with known cholesterol problems is selected from the large populations of patients. The following table gives the number of points by which each person's cholesterol level was reduced.

Table 16.10. Reduction levels of cholesterol by drugs

Pair	1	2	3	4	5	6	7	8
Drug X	29	32	31	32	32	29	31	30
Drug Y	26	27	28	27	30	26	33	36

Test whether there is any significant difference between the mean reductions of cholesterol levels by two drugs at 1% level of significance. Find 99% confidence interval for the difference between the population means.

Solution. (i) Let the observations have been selected from a bi-variate normal population. So, it is necessary to use paired t -test for testing the difference between the mean reduction levels. Let us formulate the null and alternative hypothesis as

$$H_0 : \mu_X = \mu_Y \quad \text{against} \quad H_1 : \mu_X \neq \mu_Y.$$

The appropriate test statistic is: $t = \frac{\bar{d}}{se(\bar{d})} \sim t_{n-1}$, which is distributed as t with $n - 1$ df,

where, $\bar{d} = \frac{\sum d}{n}$, $s_d^2 = \frac{\sum (d - \bar{d})^2}{n - 1}$ and $se(\bar{d}) = \frac{s_d}{\sqrt{n}}$.

Here, $n = 8$, degrees of freedom is $n - 1 = 7$.

It is two-tailed test, so at 1% level of significance the critical region is

$$|t| > t_{n-1, \alpha/2} = t_{7, 0.005} = 3.499.$$

Now, let us construct the following table for calculation of mean and standard deviation of d .

Table 16.11. Calculation of mean and standard deviation of d

Pair	1	2	3	4	5	6	7	8
Drug X	29	32	31	32	32	29	31	30
Drug Y	26	27	28	27	30	26	33	36
$d = X - Y$	3	5	3	5	2	3	-2	-6
$(d - \bar{d})$	1.38	3.38	1.38	3.38	0.38	1.38	-3.63	-7.63
$(d - \bar{d})^2$	1.89	11.39	1.89	11.39	0.14	1.89	13.14	58.14

We have $\bar{d} = 1.625$, $s_d^2 = 14.27$, $s_d = 3.777$

$$\text{So, } t = \frac{\bar{d}}{se(\bar{d})} = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{1.625}{3.777/\sqrt{8}} = 1.22.$$

- **Decision.** The observed value of t is smaller than absolute value of critical value, that means does not fall in the critical region, so we fail to reject null hypothesis.
- **Conclusion.** There is no significant difference between the average reduction of cholesterol level by two drugs, that means, drug X and drug Y are equally effective.

(ii) The 99% confidence interval for the difference of population average reduction of cholesterol levels is given by

$$\bar{d} - t_{n-1, \alpha/2} \cdot se(\bar{d}) < \mu_X - \mu_Y < \bar{d} + t_{n-1, \alpha/2} \cdot se(\bar{d}) = (-3.05, 6.30).$$

Example 16.7.8. (Matched observations or paired sample) Ten persons were appointed as probationary officers in an office. Their performance was noted by taking a test and the marks were recorded out of 100. After training for a 6-months period, another test was conducted. The marks obtained by the officers before and after training were as follows.

Table 16.12. Marks of employees before and after training

Employees	A	B	C	D	E	F	G	H	I	J
Before training	80	76	92	60	70	56	74	56	70	56
After training	84	70	96	80	70	52	84	72	72	50

Were the employees benefited by the training?

Solution. It would be proved that the employees were not benefited by the training if there is no significant difference between the average score obtained by them before and after training. On the other hand, it would be proved to be benefited if the average score obtained by employees significantly improved after training.

So we have to conduct a one tailed test defined by the null and alternative hypothesis as

$$H_0 : \mu_a = \mu_b \quad \text{against} \quad H_1 : \mu_a < \mu_b.$$

The appropriate test statistic is $t = \frac{\bar{d}}{se(\bar{d})} \sim t_{n-1}$, which is distributed as t with $n - 1$ df,

$$\text{where, } \bar{d} = \frac{\sum d}{n}, \quad s_d^2 = \frac{\sum (d - \bar{d})^2}{n - 1} \quad \text{and} \quad se(\bar{d}) = \frac{s_d}{\sqrt{n}}.$$

Here, $n = 10$, degrees of freedom is $n - 1 = 9$.

It is one-tailed test, so at 5% level of significance the critical region is

$$t < -t_{n-1, \alpha} = -t_{9, 0.05} = -1.833.$$

Now, let us construct the following table for calculation of mean and standard deviation of d